B.TECH. (CSE)

V SEMESTER

UE20303 –SOFTWARE ENGINEERING

PROJECT REPORT

ON

CONFERENCE ROOM BOOKING SYSTEM

SUBMITTED BY,

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PROPOSAL OF THE PROJECT

PROJECT TITLE : CONFERENCE ROOM BOOKING SYSTEM

In every organization there is always need of meeting and conference rooms, to conduct various events. It is found that there is one conference hall in every organization, whether it is an educational institution or any company. Many different departments have to share this single conference hall for conducting its event. Hence there is always a possibility of the hall being booked by two or more departments on the same day. The clash in timing will be known to the departments only when the day of the event has reached, by that time it will be too late and very little time left for alternate arrangement. Hence an efficient and user friendly system is required to reserve the hall beforehand and make the information available to others to check the status of the hall before booking.

SOFTWARE REQUIREMENTS SPECIFICATION

This section illustrates the functional requirements for CRRS web interface which is having 5 different web pages that enables customer to select and reserve room online. Each web page issupported by set of specific requirements given in below tables.

The requirements below are developed within the scope from customer perspective only.

Checking Availability Page :

Requirement Shell

VER.	Version of requirement
REQ ID	Unique identification code of requirement
DESCRIPTION	Description of requirement
PRIORITIZATION ATTRIBUTE	Prioritization of requirement according to prioritization plot
TRACEABILITY ATTRIBUTE	Traceability relation to other artefacts

Requirement Shell in the table :

Ver.	Req ID	Description	PrioritizationAttribute	TraceabilityAttribute
V1	R1.1	This page shall have a field for arrival and departuretime selection	High	
V1	R1.2	This page shall have an option for selecting roomtype	High	Content of R1.3
V1	R1.3	Page shall show check availability and rates buttonto redirect customer to select a room page	High	Precondition to the requirements fromSYSF2

Selection of a Room Page :

Ver.	Req ID	Description	Prioritizatio nAttribute	Traceabilit yAttribute
V1	R2.1	Check the availability of rooms	High	Precondition to R2.2
V1	R2.2	Page shall display available rooms according to R2.1	High	Evolution of R2.1
V1	R2.2.1	Page shall display room pictures of all types of available rooms	Medium	Content R2.2
V1	R2.2.2	Page shall display room prices per hour of all available rooms	High	Content of R2.2
V1	R2.2.3*	Page shall display select & continue button which will redirect the customer to customer details page	High	Precondition to requirements to SYSF3
V1	R2.3	Page shall have sorting function to sort by price	High	
V1	R2.4	Price shall be displayed in Rupees	High	Miscellaneous of R2.2.2
V1	R2.5	Page shall display button to previous page with previous page name on the button	High	
V1	R2.6	Page shall display cancel button which will redirect the customer to Conference Room home page	High	

Customer Details Page :

Ver.	Req ID	Description	Prioritizatio nAttribute	Traceabilit yAttribute
V1	R3.1*	Page shall display text field for Firstname	High	
V1	R3.2*	Page shall display text field for Lastname	High	
V1	R3.3*	Page shall display field for Telephone number	Medium	
V1	R3.4*	Page shall display field for E-mail address	High	
V1	R3.5*	Page shall display mandatory checkbox fieldfor agreement of terms and conditions	High	
V1	R3.6	Page shall display submit details button whichwill redirect to payment page	High	Precondition to requirements ofSYSF4
V1	R3.7	Page shall display button to previous pagewith previous page name on the button	High	
V1	R3.8	Page shall display cancel button which willredirect the customer to Conference Room home page	High	

Payment Page :

Ver.	Req ID	Description	Prioritizatio nAttribute	Traceabilit yAttribute
V1	R4.1	Page shall display Total price of selected room forselected number of nights	High	Evolution of R2.2.2and R1.1
V1	R4.2	Page shall display field for Credit card number	High	
V1	R4.4	Page shall display field for expiration date ofcredit card	High	
V1	R4.5*	Page shall display mandatory checkbox field foragreement of terms and conditions	High	
V1	R4.6	Page shall display a Note: Card details will beused for authentication purposes only and the payment will charged upon customer check-in	High	Miscellaneous ofR4.7
V1	R4.7	Page shall display continue button which willredirect to confirmation page	High	Precondition to requirements ofSYSF5
V1	R4.8	Page shall display button to previous page withprevious page name on the button	High	
V1	R4.9	Page shall display cancel button which willredirect the customer to Conference Room home page	High	

Confirmation Page :

Ver.	Req ID	Description	Prioritizatio nAttribute	Traceabilit yAttribute
V1	R5.1	Page shall display previous inputs by customeras SYSF3R3.1-SYSF3R3.4	High	Evolution of SYSF3R3.1 - SYSF3R3.4
V1	R5.2	Page shall display arrival and departure times according to customer selection in SYSF1R1.1	High	Evolution of SYSF1R1.1
V1	R5.3	Page shall display room selection details chosen by customer according SYSFR2.2.3	High	Evolution of SYSFR2.2.3
V1	R5.4	Page shall display total price according toSYSF4R4.1	High	Evolution of SYSF4R4.1

V1	R5.5	Page shall display customer payment details entered by customer according to SYSF4R4.2- SYSF4R4.4	High	Evolution of SYSF4R4. 2- SYSF4R4. 4
V1	R5.6	Page shall display confirm button which will redirect to Home page	High	
V1	R5.6.1	Page shall display a pop-up with a message: Thank you for your booking		
V1	R5.7	When customer clicks confirm button, All details from confirmation page will be stored in the Conference Room 's common database.	High	
V1	R5.8	Page shall display button to previous page with previous page name on the button	High	
V1	R5.9	Page shall display cancel button which will redirect the customer to Conference Room home page	High	
VI	R5.10	Upon Confirmation "Thank you for your booking" message shall be displayed	Medium	

Other non-functional Requirements :

This section illustrates the non-functional requirements which includes performance, reliability,security, maintainability, portability requirements, use cases, use-case scenarios, solution- oriented requirements modelling, software quality attributes for CRRS web interface.

Performance Requirements :

Ver.	Req ID	Description	PrioritizationAttribute	TraceabilityAttribute
V1	PFR1	Every booking submission & modificationshould be updated in Conference Room System's common database within 10 seconds after each submission & modification activity	High	Constraints to R5.7

V1	PFR2	Results for cross checking of availabilityof rooms in internal database and customer's choice shall be in 5 seconds	High	Constraints to R2.1
V1	PFR3	Webpage UI load time should be within 2secs	High	Constraints to R1.6,R2.2.3, R2.5, R2.6, R3.6, R3.7, R3.8, R4.7, R4.8, R4.9, R5.9, R5.10, R5.6
V1	PFR4	Redirection page load time shall be within2 secs	High	Constraints to R1.6,R2.2.3, R2.5, R2.6, R3.6, R3.7, R3.8, R4.7, R4.8, R4.9, R5.9, R5.10, R5.6

Reliability Requirements :

Ver.	Req ID	Description	Prioritizatio nAttribute	Traceabilit yAttribute
V1	RR1	Database should be synchronized to cloudevery 5 minutes	High	
V3	RR2	System shall be accessible 98% of the time	High	

Maintainability Requirements :

Ver.	Req ID	Description	Prioritizatio nAttribute	Traceability Attribute
V3	MTR1	No method in any object may exceed 100lines of code	High	
V1	MTR2	System failure recovery shall be within 40mins	High	

Portability Requirements :

Ver.	Req ID	Description	Prioritizatio nAttribute	Traceability Attribute
V3	PTR1	System shall be accessible on mobile devices	High	Miscellaneous
V3	PTR2	System shall be accessible on Google Chrome,Firefox, Safari, Opera and Internet explorer	High	Miscellaneous

PROJECT PLAN

1. Lifecycle :

The Software Development Life cycle followed will be AGILE specifically SCRUM since we will be performing tests, taking feedback and refining our product often according to the test results, and as per the feedback received from the stakeholders.

2. Tools :

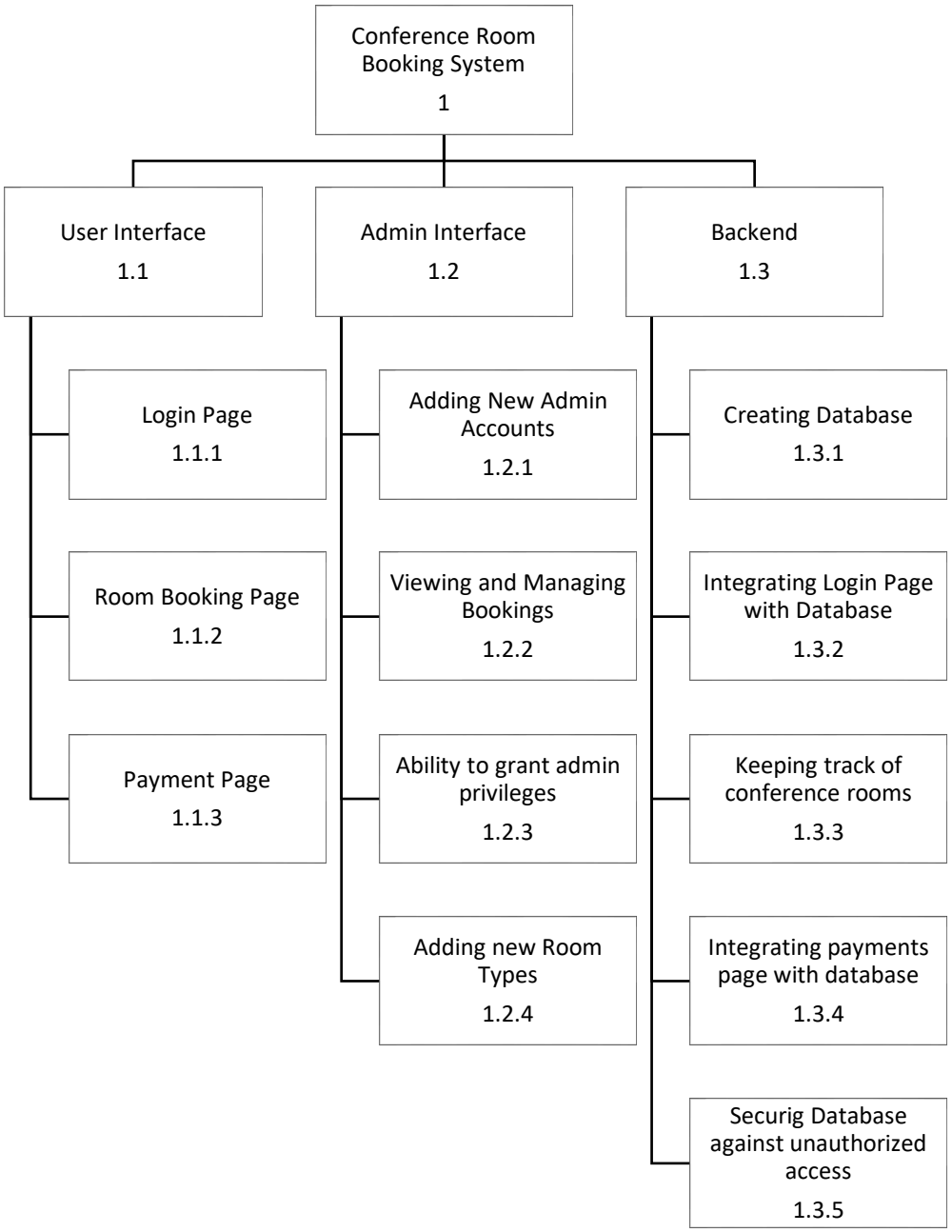
- Planning Tool : Gantt Chart
- Design Tool : Figma
- Version Control Tool : Git and GitHub
- Development Tool : VS Code IDE and GitHub
- Bug Tracking Tool : Zoho bug tracker

3. Deliverables :

PROJECT NAME	Conference Room Management System			PROJECT MANAGER	
PROJECT DELIVERABLE	A fully working conference room management system where users are able to book rooms and admin is able to manage conference rooms				
SCOPE STATEMENT	User should be able to login and book rooms while admin should be able to add rooms, manage bookings and create new admin accounts				
START DATE	24/08/2022	END DATE	20/11/2022	OVERALL PROGRESS	100%

TASK NAME	ASSIGNED TO	START DATE	END DATE	DURATION in days
SPRINT 1		24/08	20/09	28
Building Login Page frontend	Sanath Sreekanth	24/08	30/08	7
Creating database and integrating login page frontend with database	Sanmat	31/08	13/09	14
Building security features to secure database against unauthorized access	Shamith V S	13/09	20/09	7
SPRINT 2		21/09	18/10	28
Displaying different types of conference rooms available	Sampreeth Nadig	21/09	27/09	7
Building Room booking page and integrating with backend	Shamith V S	21/09	04/10	14
Building Payments page and integrating with backend	Sanath Sreekanth	05/10	18/10	14
SPRINT 3		19/10	16/11	28
Enabling manager to add new manager accounts with manager privileges and storing new account information in database	Sanmat	19/10	25/10	7
Enabling manager to add new room types	Sampreeth Nadig	25/10	09/11	14
Creating manager dashboard to manage bookings	Shamith V S	02/11	16/11	14

4. Work breakdown structure :

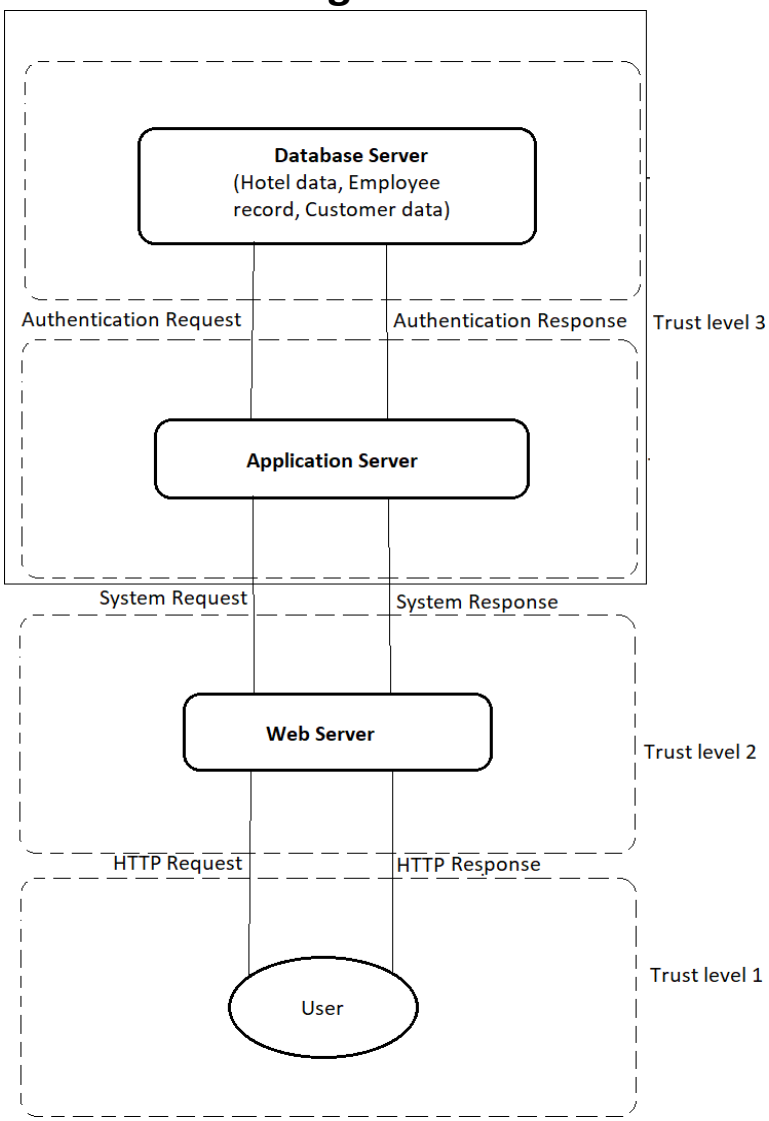


5. Gantt Chart :

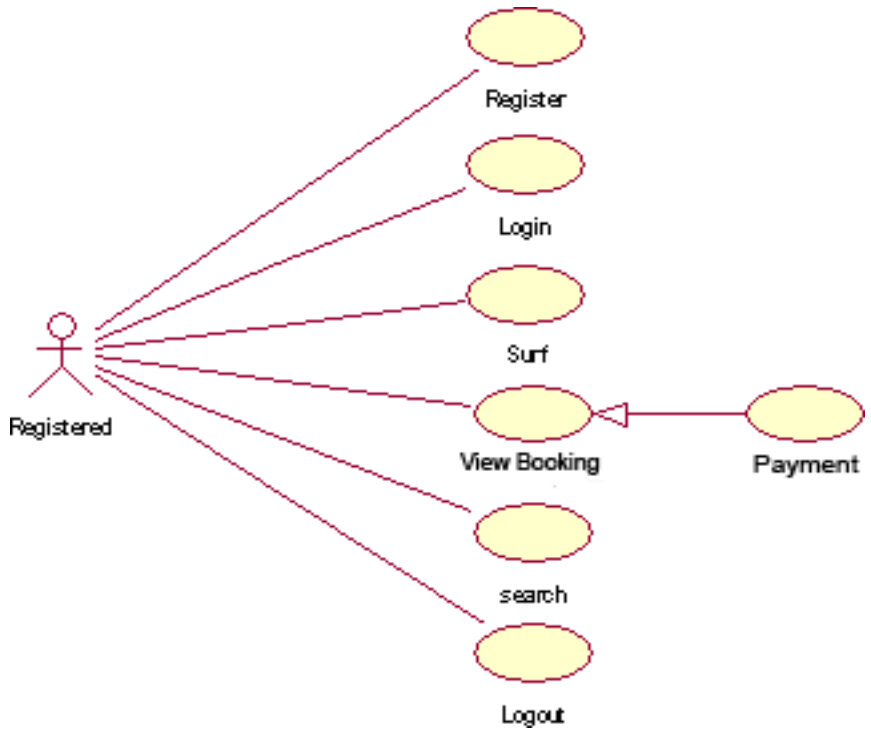
TaskID	Task Name	Start Week	End Week		WK01	WK02	WK03	WK04	WK05	WK06	WK07	WK08	WK09	WK10	WK11	WK12	WK13	WK14	WK15
01	**SPRINT1**	WK01	WK04		■	■	■	■											
01.01	Login Page	WK01	WK01		■														
01.02	Creating database and	WK02	WK03			■	■												
01.03	Securing Database	WK04	WK04					■											
02	**SPRINT2**	WK05	WK08						■	■	■	■							
02.01	Room Availability Page	WK05	WK05						■										
02.02	Room Booking Page	WK05	WK06						■	■									
02.03	Payments Page	WK07	WK08								■	■							
03	**SPRINT3**	WK09	WK12										■	■	■	■			
03.01	Allowing managers to	WK09	WK09										■						
03.02	Allowing managers to add	WK10	WK11											■	■				
03.03	Managing bookings page	WK11	WK12												■	■			

DESIGN DIAGRAMS

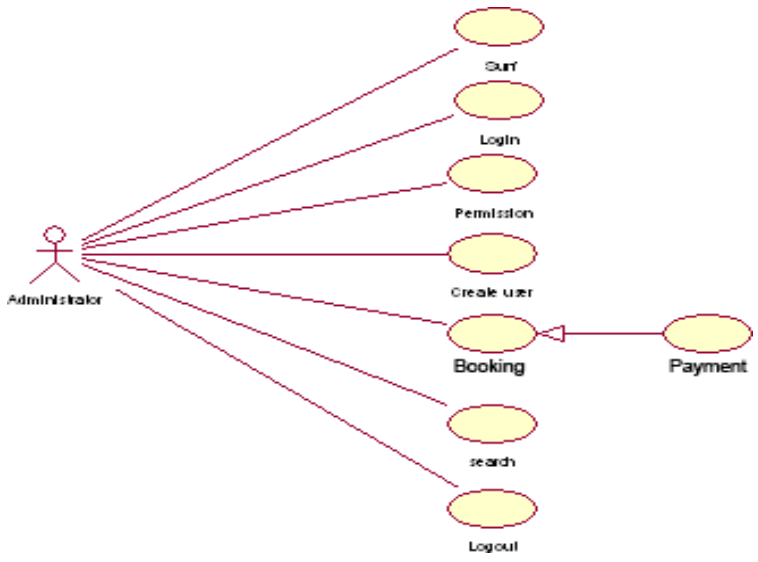
Architecture Diagram :



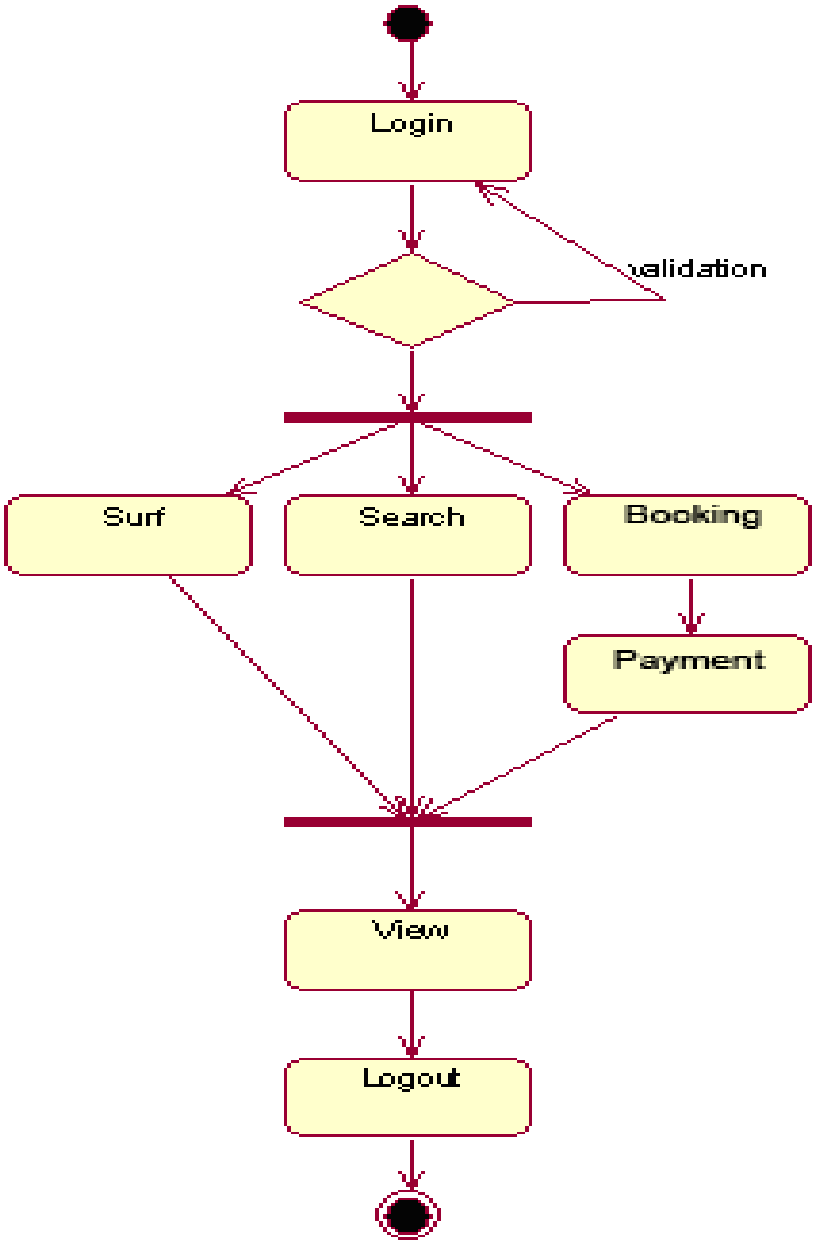
USER use case diagram :



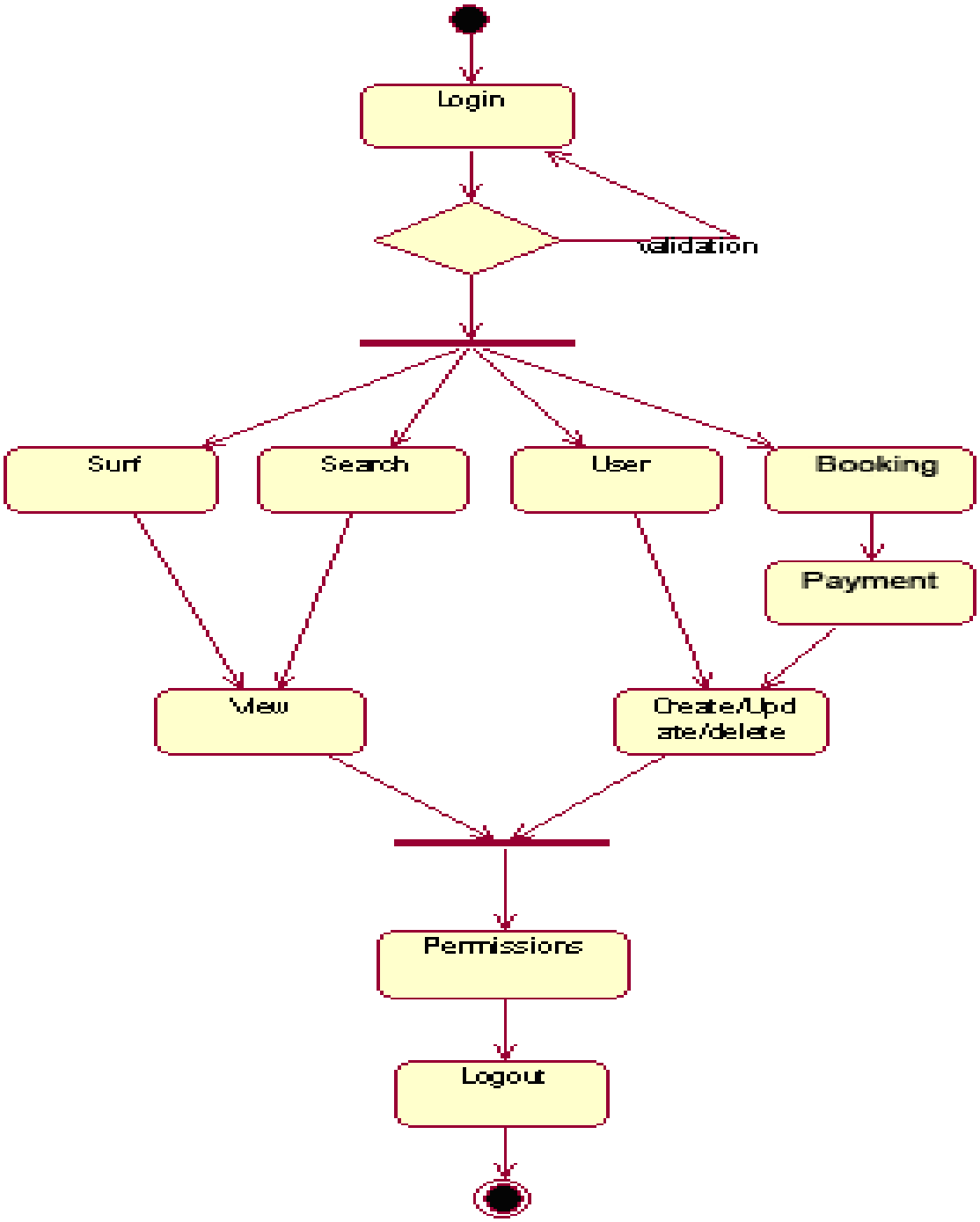
ADMIN use case diagram :



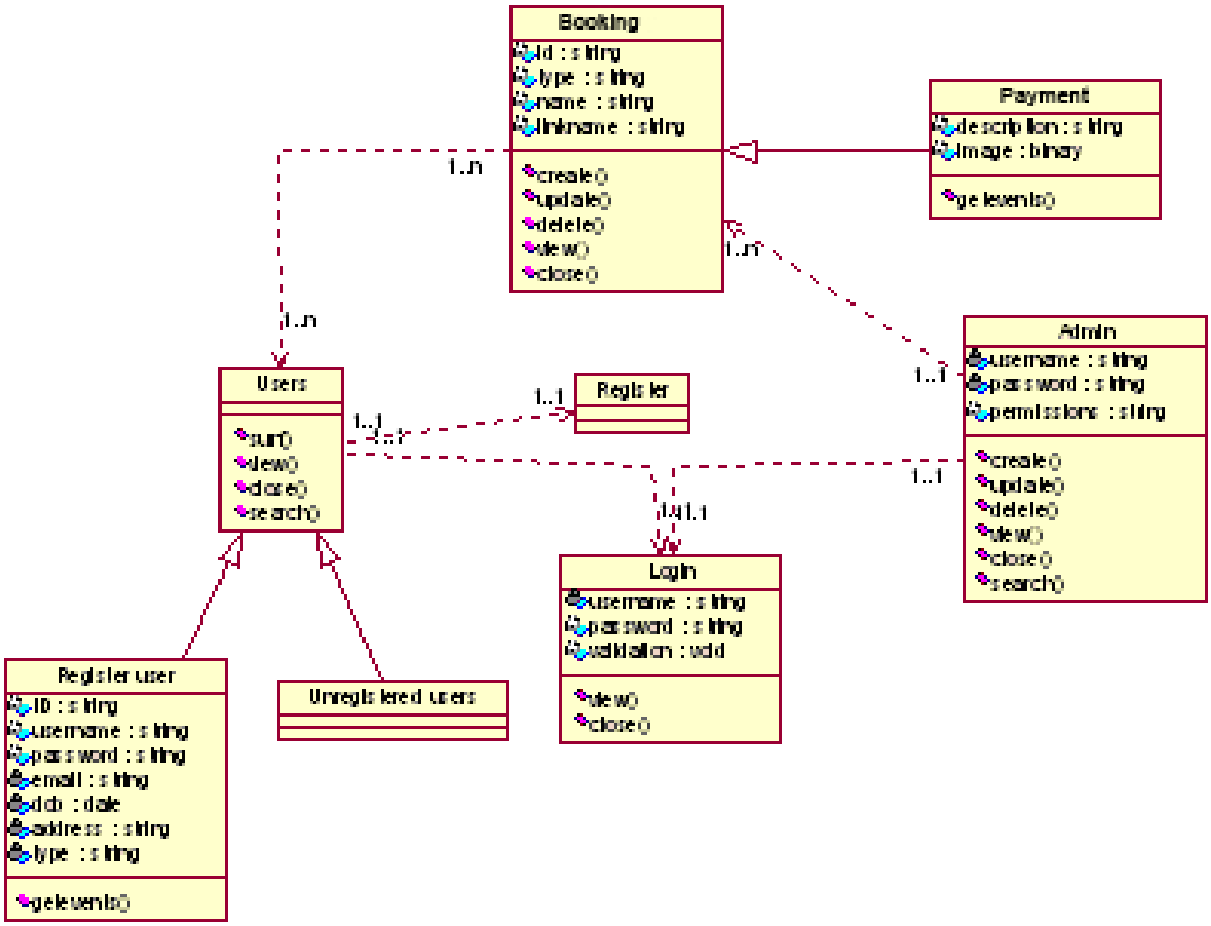
User activity diagram :



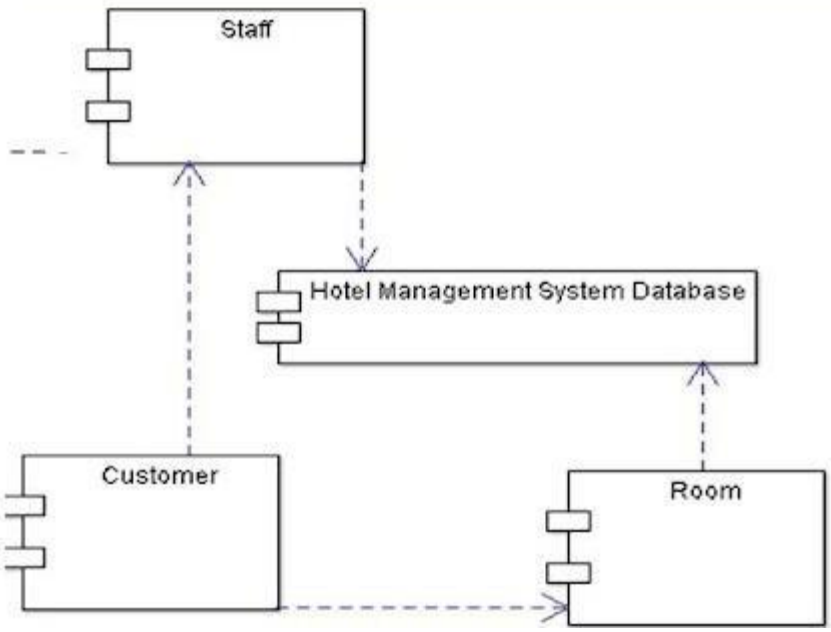
Administrator activity diagram :



Class diagram :



Component diagram :



Data flow diagram :

TEST CASES

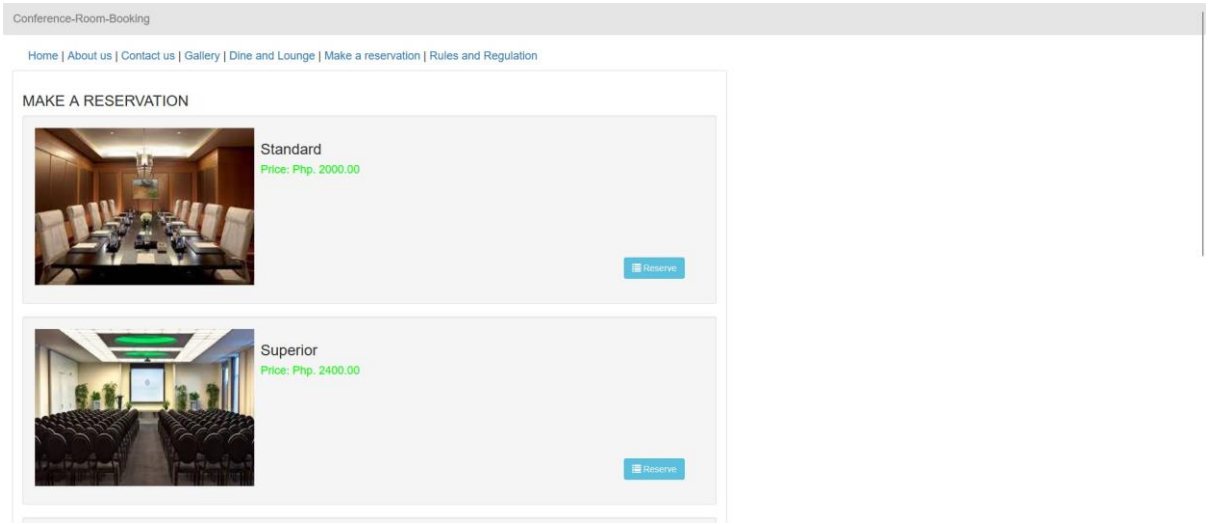
Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
DF - 1	Validate Date Field	Checking the Date of check in is greater than present date or not	Access to Internet and a browser	Check by entering a valid date in the date picker field	-	Only dates Greater than present date should be able to select	Only dates Greater than present date are valid to select	PASS
PN - 1	Validate Phone Number field	Check if the phone number is of particular format or not, and it's length also	Access to Internet and a browser	Check by entering a valid Phone number in the date picker field	Valid formats of the Phone numbers across the world	Only phone numbers of valid format should be accepted	Only phone numbers of valid format are accepted	PASS
BR - 1	Validating The back to reservation button	On pressing the button Back to reservation if the page goes to room type or not	Access to Internet and a browser	Check by pressing the back to reservation field after a room reservation	-	On pressing back to reservation, page should redirect to Room types page	On pressing back to reservation, page is redirecting to Room types page	PASS
AD - 1	Valid Admin login	On entering valid username and password admin should be able to login	Access to Internet and a browser	Check by entering valid admin username and password	Admin username and password table	On login admin should be able to see booking pending, checked ins, checked outs and past details	On login admin is be able to see booking pending, checked ins, checked outs and past details	PASS

AD-2	Check in and room number and number of rooms entry validation	In pending list on pressing check in button room number and number of days of reservation fields should appear	Access to Internet and a browser	In pending list after checking all customer details and pressing check in button all the details should be added to database and room number and number of days of reservation fields should appear	Available room numbers	Should be able to select one of the rooms from available rooms and a field to add number of days	We were able to select one of the rooms from available rooms and a field to add number of days	PASS
AD - 3	Check out validation	In check in list on clicking checkout the customer details should appear in checked out list	Access to Internet and a browser	In check in list on clicking checkout the customer details should be added to database and all the details should appear in checked out list	-	Check in and check out time should be added to database	Check in and check out time are added to database	PASS

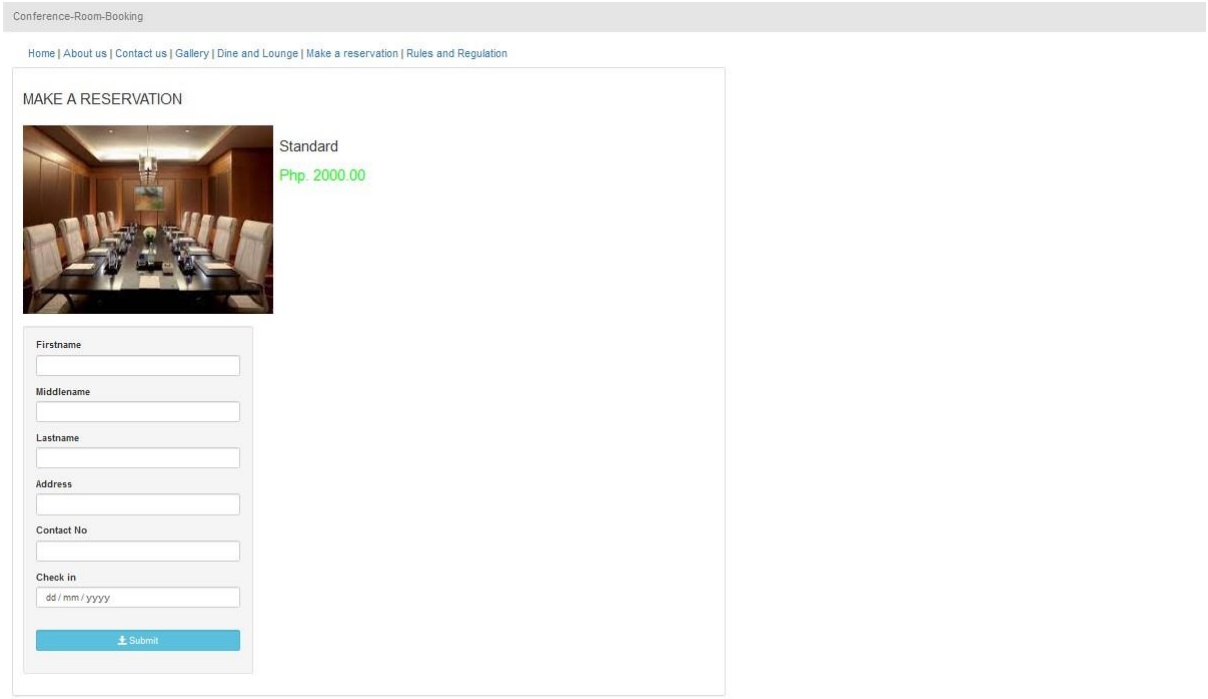
SCREEN SHOTS OF FRONT END

USER Side :

Type of rooms :

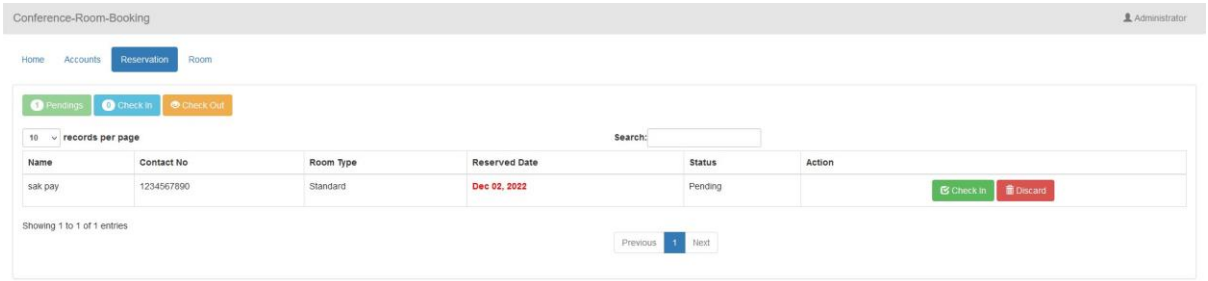


User details entry :

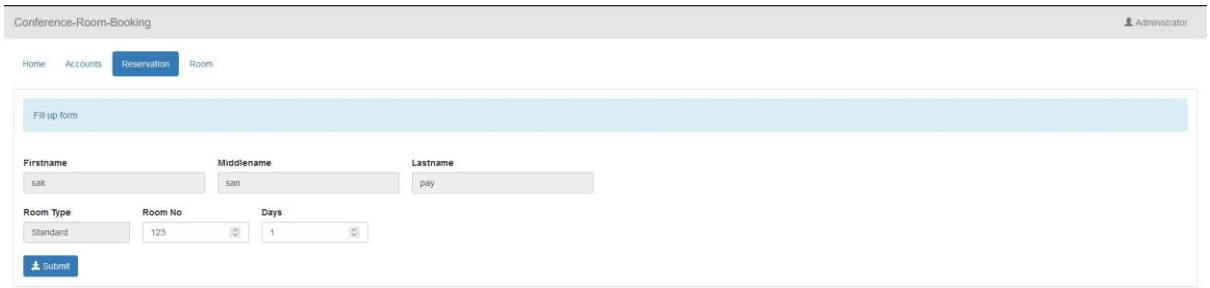


ADMIN Side :

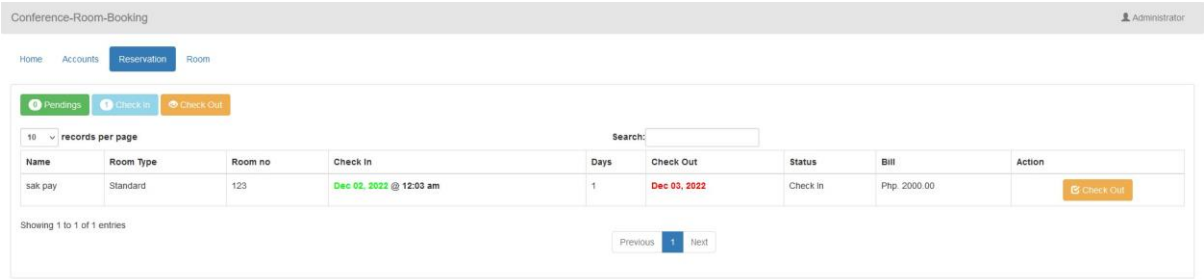
Pending check ins :



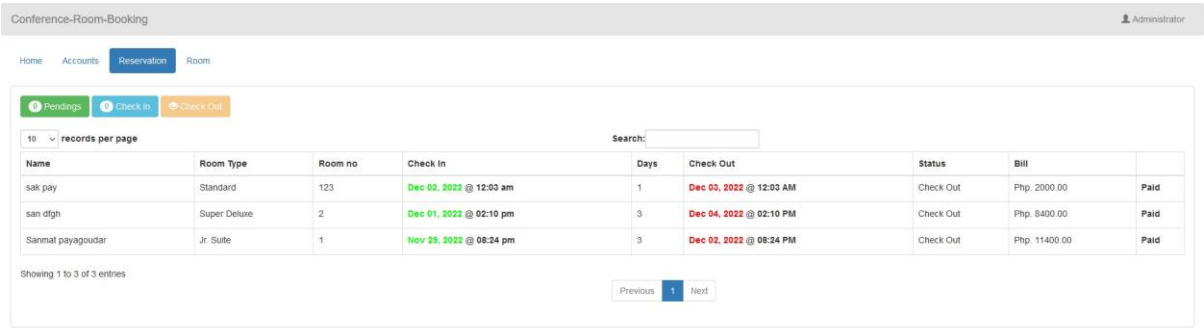
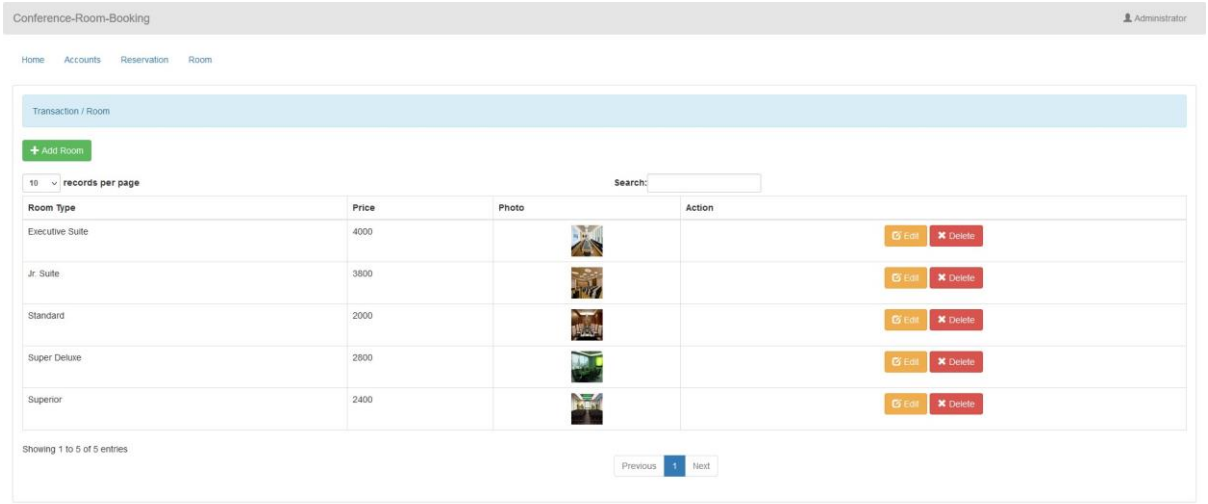
Entering Room number and number of days :



Check in :



Check out :



Details of room type :